

Optimal design of LCL filter in grid-connected inverters

ISSN 1755-4535

Received on 4th June 2018

Revised 9th November 2018

Accepted on 28th February 2019

E-First on 6th June 2019

doi: 10.1049/iet-pel.2018.5518

www.ietdl.org

Yong-Jung Kim¹, Hyosung Kim¹ ✉¹Division of Electrical, Electronic and Control Engineering, Kongju National University, 275 Budae, Cheonan, Chungnam, 310-80, Republic of Korea

✉ E-mail: hyoskim@kongju.ac.kr

Abstract: As an essential part in technologies for energy storage systems (ESSs) or renewable energy systems (RESs), grid-connected inverters need power passive filters to meet grid regulations. As typical passive filters, L filter and LCL filter are employed. Although LCL filter is more cost-effective than L filter, a design of LCL filter is more complicated than L filter. LCL filter has three filter elements: inverter-side inductor, grid-side inductor, and filter capacitor. To design the three elements for LCL filter, three or more simultaneous equations should be required, which means that three or more design target should exist. Since conventional design methods for LCL filter do not suggest clear targets and do follow try & error method which cannot be matched exactly with design targets, the conventional methods cannot be optimal design. In this study, a filter inductance ratio to minimise total filter inductance, a filter admittance to meet grid regulation and characteristic impedance for low current stress of switch stack are suggested as design goals. Optimal design equation is proposed to meet the three design goals. The proposed method can solve unique filter elements for LCL filter without iterative try & error. The design method is evaluated through simulation and experiment.

1 Introduction

Grid-connected inverters handle power exchange between DC power generated by renewable energy and AC grid. Pulse width modulation (PWM) control and dead time control are general control methods for grid-connected inverters [1]. Outputs of the inverter include high-order harmonics by PWM control and low-order harmonics by dead time control. If these harmonics flow into grids without attenuation, power quality of grid is deteriorated. In IEEE 519-2014 standards, total demand distortion (TDD) of all power generation equipment is limited as 5% [2]. Power passive filter is necessary to attenuate harmonics generated by the inverter. There are two type of passive filter for grid-connected inverter: *L* filter and *LCL* filter [3].

L filters play a role as a first order low-pass filter (LPF) to attenuate the harmonics of grid-side current. To limit TDD (%) of grid-side current below 5%, a quite large amount of filter inductance is required, which increases volume and weight of the inverter systems and makes the output dynamics of the inverter poor [4].

LCL filters play a role as a third-order LPF which is combined with *LC* filter (second order LPF) and *L* filter (first-order LPF). Although *LCL* filters are smaller and cheaper than *L* filters, resonance issues are followed [5]. Resonance caused by *LCL* filter declines output power quality of grid-connected inverters and control stability of the inverter system. Thus, it is important to decide resonance frequency in *LCL* filter design. Passive damping method can be applied to suppress resonance of *LCL* filter, but power loss is caused in damping resistor [6, 7]. Active damping method can be applied to prevent power loss of the damping resistor, but structure of the controller becomes complicated and the damping capability is limited depending on the switching frequency [8–15].

LCL filter has three filter elements: inverter-side inductor, grid-side inductor, and filter capacitor. To design the three elements for *LCL* filter, three or more simultaneous equations should be required, which means that three or more design target should exist.

Conventional design methods for *LCL* filter apply filter inductance ratio as a design target. That is because optimal filter

inductance ratio means the smallest filter reactor price. Some researchers described that symmetric inductance ratio is a cost-effective way to minimise filter inductances in same filter admittance condition. [16]. Another researchers described that bigger inverter-side inductance than grid-side inductance has advantages in reducing switching losses on inverter-side inductor and switching stress on switch stack [17]. The others described that bigger grid-side inductance than inverter-side inductance is good to reduce costs of both inductors, because core for inverter-side inductor is more expensive than grid-side inductor [18].

These discussions on filter inductance ratio give limited insights to important characteristics for *LCL* filter design, because they focused on the effects by just inverter-side current harmonics or grid-side current harmonics. Hence, optimal inductance ratio should be established through frequency response analysis of both inverter-side current harmonics and grid-side current harmonics in *LCL* filter together.

Conventional design methods for *LCL* filter have various goals: filter inductance ratio, resonance frequency, tolerable current ripple on inverter side or grid side, reactive power ratio absorbed in filter capacitor and etc. In [19], two design goals are selected: filter inductance ratio and resonance frequency. Unfortunately, the method does not suggest the way to decide resonance frequency and total filter inductance in order to limit TDD (%) of grid-side current below 5%. In [20], three design goals are selected: reactive power ratio absorbed in filter capacitor and tolerable current ripple on inverter side and grid side. However, the method does not suggest the way to decide tolerable current ripple in order to limit TDD (%) of grid-side current below 5%. In [21, 22], four design goals are selected: reactive power ratio absorbed in filter capacitor and tolerable current ripple on inverter side, resonance frequency and filter inductance ratio. The methods do not suggest the way to decide tolerable current ripple and resonance frequency of *LCL* filter.

These methods employ try & error method which iteratively changes the combination of filter elements, until the grid regulation is satisfied. Design by try & error method is time-consuming and cannot solve optimal filter elements to achieve selected design goals.

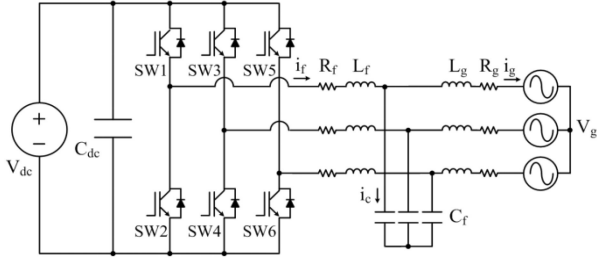


Fig. 1 Scheme of three-phase grid-connected inverter with LCL filter

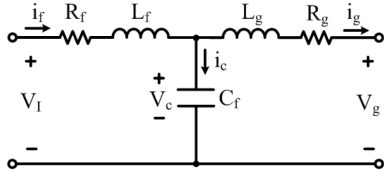


Fig. 2 Single-phase LCL filter configuration with internal resistances of filter inductors

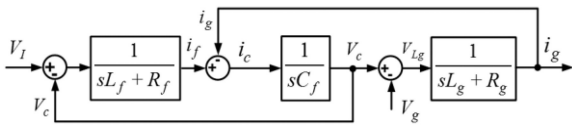


Fig. 3 LCL filter block diagram

Considered in addition to the grid regulation and cost-effectiveness for LCL filters, an important design goal is a current stress on switch stack. In L filter, inverter-side switching current is same with grid-side switching current. Yet, in LCL filter, inverter-side current ripple is higher than grid-side current ripple. That is because the filter capacitor of the LCL filter increases the switching ripple of the inverter-side current. Since current stress on switch stack by the switching current degrades a reliability of the inverter system, the stress on switch stack should be considered in LCL filter design.

Here, a filter inductance ratio to minimise total filter inductance, a filter admittance to meet grid regulation, and characteristic impedance for low current stress of switch stack are suggested as design goals. Optimal design equation is proposed to satisfy the design goals. The proposed method can solve unique filter elements for LCL filters without iterative try & error. The design method is evaluated through simulation and experiment.

2 Optimal LCL filter design

2.1 Transfer function analysis of LCL filter and L filter

A typical circuit diagram of a three-phase grid-connected inverters with LCL filter is shown in Fig. 1. In the conditions that each phase voltage of the inverters and grids is symmetric and LCL filters are balanced, three-phase systems could be transformed equivalently into single-phase systems. Fig. 2 shows single-phase LCL filter configuration with internal resistances of filter inductors. Here, grid impedance is assumed as a part of grid-side filter inductor. A block diagram of LCL filter is shown as Fig. 3. A transfer function of LCL filter for grid-side current i_g over inverter output voltage V_1 is followed as; (see (1)), where $\tau_f = L_f/R_f$, $\tau_g = L_g/R_g$. τ_f and τ_g are time constants of inverter- and grid-side filter inductors. If time constants of inverter- and grid-side filter inductors are equal to τ , the transfer function of (1) can be followed as;

$$G(s)_{LCL} = \frac{I_g(s)}{V_1(s)} = \frac{(1/(r(1-r)L^2C_f))}{(s + (1/\tau))(s^2 + 2\zeta\omega_{res}s + \omega_{res}^2)} \quad (2)$$

$$f_{res} = \frac{1}{2\pi} \sqrt{\frac{1}{r(1-r)LC_f}} \quad (3)$$

$$\zeta = \frac{1}{2\omega_{res}\tau_f} = \frac{1}{2\omega_{res}\tau_g} = \frac{1}{2\omega_{res}\tau} \quad (4)$$

Here, $L = L_f + L_g$ and $r (=L_f/L)$ is a filter inductance ratio of inverter-side filter inductor L_f against the total filter inductor L . A resonance frequency of LCL filter is followed as (3). The damping ratio of LCL filter is determined by the time constant of filter inductor and the resonance frequency of LCL filter, as shown in (4).

In the grid-connected inverters with LCL filters, switching harmonics of inverter-side current are as important as grid-side current, because switching ripples of inverter-side current result in power losses on the filter inductor and current stress on the switch stack.

A transfer function of LCL filter for inverter-side current i_f over inverter output voltage V_1 can be followed as

$$H(s)_{LCL} = \frac{I_f(s)}{V_1(s)} = \frac{1}{rL\left(s + \frac{1}{\tau}\right)} \left\{ 1 - \frac{\omega_f^2}{s^2 + 2\zeta\omega_{res}s + \omega_{res}^2} \right\} \quad (5)$$

$$\frac{V_c(s)}{V_1(s)} = \frac{\omega_f^2}{s^2 + 2\zeta\omega_{res}s + \omega_{res}^2} \quad (6)$$

where ω_f^2 is $1/rLC_f$.

To figure out the behaviours of LCL filter, comparing with L filter is helpful. A transfer function of L filter for grid-side current i_g over inverter output voltage V_1 is followed as

$$G(s)_L = \frac{I_g(s)}{V_1(s)} = \frac{1}{L\left(s + \frac{1}{\tau}\right)} \quad (7)$$

where L is the filter inductance and τ is the time constant of L filter. For comparison analysis between L filter and LCL filter, those are assumed that filter inductance of L filter is equal to total filter inductance in LCL filter and that the time constants of the filter inductors in both filters are equal to τ .

2.2 Filter inductance ratio for minimising filter inductance

Filter inductance ratio ($r = L_f/L$) is the most cost-effective design factor in LCL filter. In order to design an optimal inductance ratio, both inverter-side current harmonics and grid-side current harmonics in a LCL filter should be considered together. To analyse both side current harmonics of a LCL filter, the frequency responses of the transfer functions are calculated as follows;

$$|G(j\omega)_{LCL}| = \frac{1}{L_f L_g C_f} \sqrt{\frac{1}{\omega^2 + \frac{1}{\tau^2}}} \sqrt{\frac{1}{\omega^4 + 2(-1 + 2\zeta^2)\omega^2\omega_{res}^2 + \omega_{res}^4}} \quad (8)$$

$$|H(j\omega)_{LCL}| = \frac{1}{rL} \sqrt{\frac{1}{\omega^2 + \frac{1}{\tau^2}}} \cdot \left| 1 - \frac{\omega_f^4}{\omega^4 + 2(-1 + 2\zeta^2)\omega^2\omega_{res}^2 + \omega_{res}^4} \right| \angle \theta_c \quad (9)$$

$$\theta_c = \tan^{-1} \{ 2\zeta\omega_{res}/(\omega_{res}^2 - \omega^2) \} \quad (10)$$

$$\frac{I_g(s)}{V_1(s)} = \frac{(1/L_f L_g C_f)}{s^3 + ((1/\tau_f) + (1/\tau_g))s^2 + ((1/\tau_f \tau_g) + (L_f + L_g/L_f L_g C_f))s + (1/\tau_f L_g C_f) + (1/\tau_f L_g C_f)} \quad (1)$$

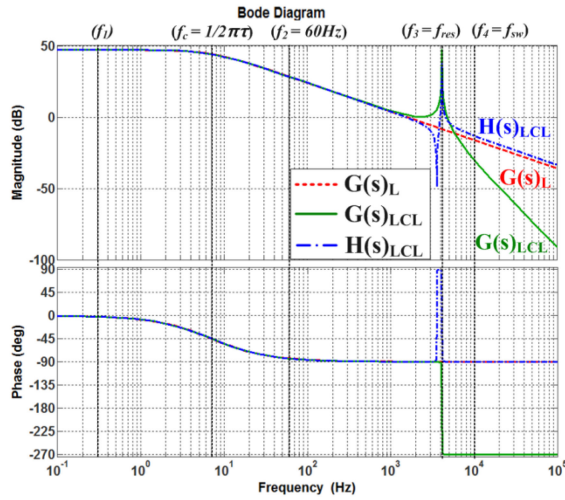


Fig. 4 Bode plot of $G(s)_L$, $G(s)_LCL$ and $H(s)_LCL$

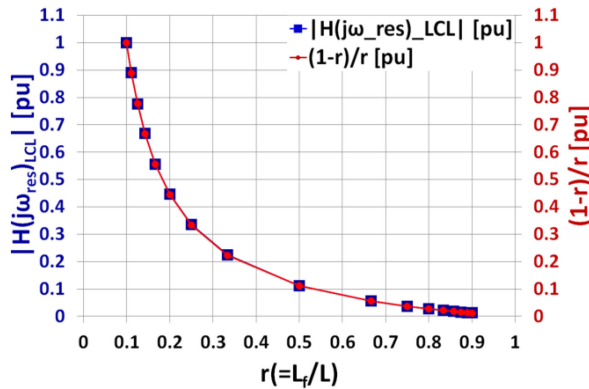


Fig. 5 Relation of $|H(j\omega_{sw})_{LCL}|$ and $(1-r)/r$ according to $r(=L_f/L)$

where θ_c is the phase difference between filter capacitor voltage V_c and inverter output voltage V_I .

The frequency response of grid-side current with L filter is followed as

$$|G(j\omega)_L| = \frac{1}{L} \sqrt{\frac{1}{\omega^2 + \frac{1}{\tau^2}}} \quad (11)$$

where ω is an angular speed, defined as 2π times frequency(f). In frequency response analyses of L filter and LCL filter, the most significant frequencies are fundamental frequency, resonance frequency, and switching frequency.

In bode plots of Fig. 4, $f_c (=1/2\pi\tau)$ is a break frequency at which frequency responses of the transfer functions changes from zero-order to first-order system. At the frequency, frequency responses of both L filter and LCL filter change from resistive characteristics to inductive characteristics. Frequency responses of both L filter and LCL filter below f_c are as follows;

$$|G(j\omega_1)_L|_{dB} = |G(j\omega_1)_{LCL}|_{dB} = |H(j\omega_1)_{LCL}|_{dB} = -20\log R \quad (12)$$

where $R = R_f + R_g$ and $\omega_1 (=2\pi f_1)$ is an angular speed far below ω_c ($f_1 \ll f_c$). Equation (12) and Fig. 4 describe that the frequency responses of L filter and LCL filter far below f_c are decided by the internal resistances of filter inductors.

The frequency response at $f_2 (=60\text{ Hz})$ indicates the dynamics of fundamental current by the inverters. Frequency responses of L filter and LCL filter at fundamental frequency f_2 are followed as

$$|G(j\omega_2)_L|_{dB} = |G(j\omega_2)_{LCL}|_{dB} = |H(j\omega_2)_{LCL}|_{dB} = -20\log\omega_2 - 20\log L \quad (13)$$

where $\omega_2 (=2\pi f_2)$ is an angular speed of the fundamental frequency (60 Hz). Fig. 4 and (13) show that dynamics of both filters are same, when both filters have same filter inductance. Around $f_2 (=60\text{ Hz})$, the slopes of bode gain in both L filter and LCL filter are equal to -20 [dB/decade] .

Around the resonance frequency $f_3 (=f_{res})$, the frequency responses of L filter and LCL filter become completely different as follows

$$|G(j\omega_3)_L|_{dB} = -20\log\omega_3 - 20\log L \quad (14)$$

$$|G(j\omega_3)_{LCL}|_{dB} = -20\log\omega_3 - 20\log L + 20\log 2\zeta \quad (15)$$

$$|H(j\omega_3)_{LCL}|_{dB} = -20\log\omega_3 - 20\log L + 20\log 2\zeta + 20\log \frac{1-r}{r} \quad (16)$$

where $\omega_3 (=2\pi f_3)$ is an angular speed of the resonance frequency with LCL filter. Since L filter is first-order system, its frequency response at the resonance frequency is decided by filter inductance and an angular speed of the resonance frequency. Compared with L filter, resonance frequency response of grid-side current in LCL filter is affected by damping ratio ζ as (15). If the damping ratio defined in (4) is substituted to (15), it can be shown that the response of grid-side current in LCL filter is determined by the internal resistance of total filter inductor.

Compared with resonance frequency response of grid-side current, resonance frequency response of inverter-side current depends on the inductance ratio r . Fig. 5 shows the relationship between $|H(j\omega_{res})_{LCL}|$ and $(1-r)/r$ when a range of the inductance ratio r is from 0.1 to 0.9. As shown in Fig. 5, resonance frequency response of inverter-side current in LCL filter is inversely proportional to the inductance ratio r .

When the inductance ratio is 0.5 in (16), the decibel value of $(1-r)/r$ is 0 dB. When the inductance ratio is smaller than 0.5 in (16), the decibel value of $(1-r)/r$ is bigger than 0 dB. It means that inverter-side resonance current is amplified, when inverter-side inductance are smaller than grid-side inductance. The increase of resonance current can degrade the stability of inverter system. When the inductance ratio is bigger than 0.5 in (16), the decibel value of $(1-r)/r$ is smaller than 0 dB. It means that inverter-side resonance current is attenuated when inverter-side inductance are larger than grid-side inductance.

Finally, frequency responses of L filter and LCL filter at switching frequency $f_4 (=f_{sw})$ are followed as

$$|G(j\omega_4)_L|_{dB} = -20\log\omega_4 - 20\log L \quad (17)$$

$$|G(j\omega_4)_{LCL}|_{dB} \approx -60\log\omega_4 - 20\log L - 20\log r(1-r)LC_f \quad (18)$$

(see (19))

$$|H(j\omega_4)_{LCL}|_{dB} \approx -20\log\omega_4 - 20\log L - 20\log r \quad (20)$$

where $\omega_4 (=2\pi f_4)$ is an angular speed of switching frequency. Switching frequency response of L filter is decided by filter

$$\begin{aligned} |H(j\omega_4)_{LCL}|_{dB} &= -20\log \left\{ \frac{1}{\omega_4 r L} \left(1 + \left| \frac{1/r LC_f}{-\omega_4^2 + j(1/\tau)\omega_4 + (1/(r(1-r)LC_f))} \right| \right) \right\} \\ &= -20\log \left(\frac{1}{\omega_4 r L} + \frac{1}{\omega_4 r L} \sqrt{\frac{\omega_f^4}{\omega_4^4 + 2(1+2\zeta^2)\omega_4^2\omega_{res}^2 + \omega_{res}^4}} \right) \end{aligned} \quad (19)$$

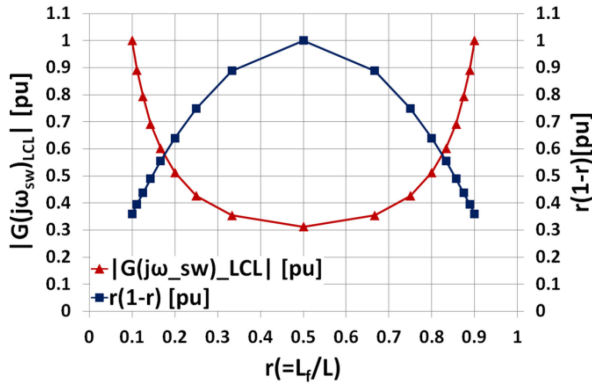


Fig. 6 Relation of $|G(j\omega_{sw})_{LCL}|$ and $r(1-r)$ according to $r(=L_f/L)$

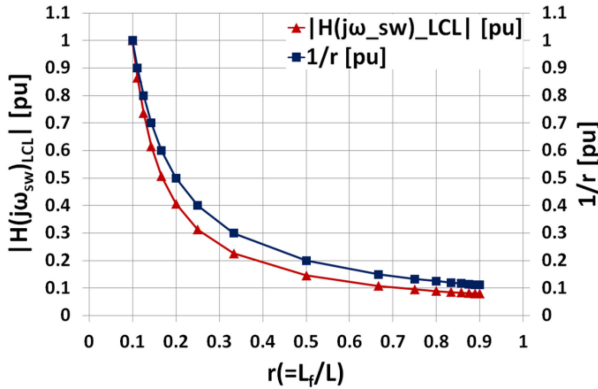


Fig. 7 Relation of $|H(j\omega_{sw})_{LCL}|$ and $1/r$ according to $r(=L_f/L)$

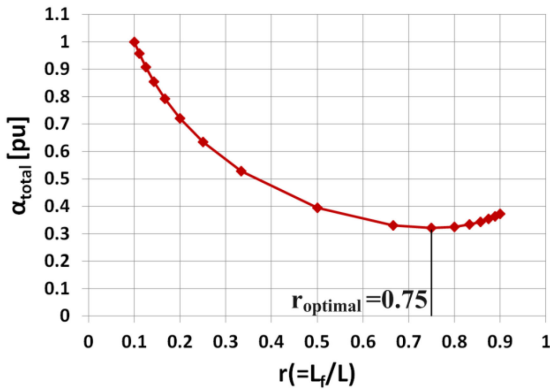


Fig. 8 Relation of α_{total} according to $r(=L_f/L)$

inductance and an angular speed of the switching frequency, like (13), (14), and (17). The slopes of bode gain in L filter are equal to -20 [dB/decade].

Switching frequency response of grid-side current in LCL filter is decided by an angular speed of switching frequency and three filter elements: filter capacitance, total filter inductance, and the inductance ratio. The slope differences of bode gain between L filter and LCL filter are -40 [dB/decade]. It physically means that the attenuation capability of switching harmonic in LCL filter is 100 times larger than L filter. Moreover, switching frequency response of grid-side current in LCL filter is inversely proportional to a multiplication of $r(1-r)$ by the inductance ratio, total filter inductance, and filter capacitance. It means that switching harmonics of grid-side current are most effectively reduced when $r(1-r)$ is maximised.

Fig. 6 shows the relationship between $|G(j\omega_{sw})_{LCL}|$ and $r(1-r)$ when a range of the inductance ratio, r is from 0.1 to 0.9. When the inductance ratio r is 0.5, the value of $r(1-r)$ is maximised and switching frequency response of grid-side current in the LCL filter is minimised. Therefore, when the inductance ratio r is 0.5, switching harmonics of grid-side current can be minimised. If the

inverter is driven by ideal PWM switching control, the optimal inductance ratio can be 0.5.

But, in the practical inverter, PWM switching control and dead time control are applied together. Even though dead time control has an advantage to prevent arm short on switch pole, the control has a fatal disadvantage to distort output voltage and current in the inverter. The distortion by dead time control causes low-order harmonics and increases TDD (%) of grid-side current [23, 24]. In aspect of grid-side current, attenuating low-order harmonics by dead time control is as important as attenuating switching harmonics by PWM control.

Since distortion of the output voltage by dead time control is inevitable in the inverter, the effective way to minimise low-order harmonics of grid-side current is to attenuate distorted switching current on inverter side. The way has two effects. One is to reduce distorted switching current flowing directly from inverter-side inductor to grid-side inductor. The other is to reduce distortion of filter capacitor voltage by decreasing distorted switching current flowing in filter capacitor. When the distortion of capacitor voltage is decreased in case that grid voltage is pure sinusoidal, distortion of grid-side current is also reduced.

Fig. 7 shows the relationship between $|H(j\omega_{sw})_{LCL}|$ and $(1/r)$ when a range of the inductance ratio r is from 0.1 to 0.9. As shown in Fig. 7, switching frequency response of inverter-side current in LCL filter is inversely proportional to the inductance ratio r . It means that the larger the inductance ratio, the smaller distorted switching current on inverter side.

Equation (19) and (20) show the increase of the inductance ratio has two effects to reduce inverter-side distorted switching current. As shown in (20), one is to decrease switching frequency response of inverter-side current, which means that inverter-side distorted switching current is reduced.

As shown in (19), the other is to decrease switching frequency response of filter capacitor voltage. It means that distortion of filter capacitor voltage is reduced. The remarkable point in (19) is that the switching harmonics of filter capacitor voltage is major harmonics source of inverter-side current as well as grid-side current. It means that the distortion of inverter-side current and grid-side current is reduced together, when distortion of filter capacitor voltage is reduced.

Thus, to design the optimal inductance ratio which minimises the harmonics generated by PWM switching control and the dead time control, switching frequency response of both side currents over filter capacitor voltage should be analysed.

$$\alpha_{ig} = 20 \log \left| \frac{(1/(r(1-r)L^2C_f))}{j\omega_{sw}(\omega_{sw}^2 + j2\zeta\omega_{res}\omega_{sw} + \omega_{res}^2)} \right| \quad (21)$$

$$\alpha_{if} = 20 \log \left| \frac{(1/r^2L^2C_f)}{j\omega_{sw}(\omega_{sw}^2 + j2\zeta\omega_{res}\omega_{sw} + \omega_{res}^2)} \right| \quad (22)$$

In (21), α_{ig} is a switching frequency response of the grid-side current i_g over filter capacitor voltage V_c . Besides, in (22), α_{if} is a switching frequency response of the inverter-side current i_f over filter capacitor voltage V_c . If common part in α_{ig} and α_{if} is ignored, core part of α_{ig} and α_{if} are followed as

$$\alpha_{ig_core} = -20 \log r(1-r) \quad (23)$$

$$\alpha_{if_core} = -20 \log r^2 \quad (24)$$

Fig. 8 shows a per unit graph for a sum of α_{if_core} and α_{ig_core} with regard to inductance ratio. Considered with distortions on grid-side current by PWM switching and dead time effect, optimal inductance ratio, $r_{optimal}$ is 0.75.

To evaluate the optimal inductance ratio, MATLAB simulations was performed. Simulation conditions follow Table 1. It is assumed that grid voltage and DC source voltage are ideal without any distortion. To examine sensitivity of the optimal inductance ratio according to filter capacitor, two capacitor conditions are applied:

filter capacitance of Case I is 3% against base capacitance and filter capacitance of Case II is 5% against base capacitance.

Fig. 9 shows graphs for THD of grid-side current in variations of the inductance ratio and percentage dead time. Fig. 9 (a) shows graphs for THD of grid-side current in filter capacitance condition of Case I. Fig. 9 (b) shows graphs for THD of grid-side current in filter capacitance condition of Case II. As shown in Fig. 9 (a) and Fig. 9 (b), under ideal PWM control conditions, the optimal inductance ratio can be 0.5, but the optimal inductance ratio converges to 0.75 as the dead time increases.

2.3 TDD limitation for grid side current

Non-linear loads lead to current harmonics that cause increased losses and heating in transformers. Power electronic converters act as non-linear loads to produce current harmonics. Specific standards such as IEEE 519 suggest guidelines for power systems against non-linear loads [2]. In the standards, total demand distortion (TDD) of all power generation equipment is limited as 5%.

Table 1 Simulation conditions for optimal inductance ratio in LCL filter

Description	Symbol	Value
Rated Power	P_{rated}	30kW
Grid voltage (phase to phase)	V_g	220Vrms
DC link voltage	V_{dc}	400V
Switching frequency	f_{sw}	10kHz
%Dead time	T_{dead}/T_{sw}	0%~3%
LCL filter	Symbol	Case I Case II
Total inductance	L	100uH 100uH
Inductance ratio	r	0.1~0.9 0.1~0.9
Filter capacitance	C_f	42.3uF 82.2uH

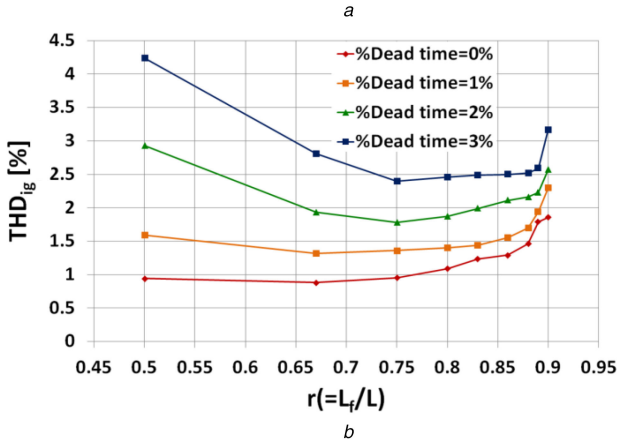
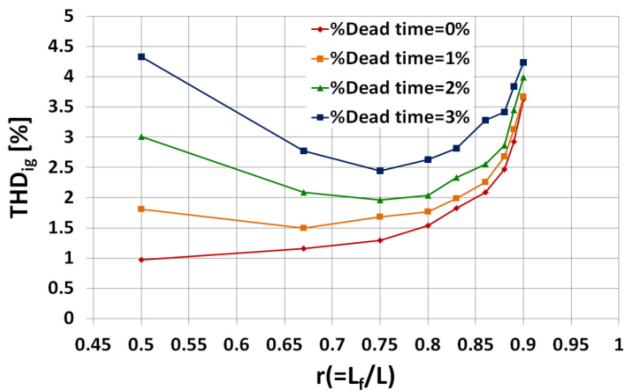


Fig. 9 Graphs for THD of i_g in variations of the inductance ratio and %dead time.

(a) Case I. (b) Case II

$$TDD = \frac{\sqrt{\sum_{k=2}^{50} I_k^2}}{I_L} \quad (25)$$

where I_L is maximum demand load current of the power generation equipment at full load condition.

The harmonics to distorted grid-side current is decided by output voltage of inverter and filter admittance between grid-side current and the output voltage. As TDD (%) depends on the harmonics at switching frequency, TDD (%) can be limited by restricting the harmonics at switching frequency.

Here, the simulation is performed to analyse the relationship between TDD (%) and filter admittance at switching frequency. It is assumed that grid-voltage and DC source voltage are ideal without any distortion. L filter was selected to simply analyse the relationship between TDD (%) and filter admittance at switching frequency. The system parameters of the grid-connected inverter are shown in Table 2. Simulation results were analysed in two switching frequency conditions; 3 kHz and 10 kHz.

Base impedance of the grid-connected inverter is followed as (26). Admittance of L filter at switching frequency is followed as (27).

$$R_{base} = \frac{V_{LL-rms}^2}{P_{rate}} \quad (26)$$

$$\left| \frac{I_g(j\omega_{sw})}{V_f(j\omega_{sw})} \right| = |Y(j\omega_{sw})| = \frac{1}{|Z(j\omega_{sw})|} = \frac{1}{\omega_{sw}L} \quad (27)$$

Fig. 10 shows THD of grid-side currents in variation of the filter admittance at switching frequency. The x-axis of Fig. 10 is the product of the filter admittance at switching frequency and the base impedance, which physically means the ratio of the filter impedance at switching frequency and the base impedance. As shown in Fig. 10, TDD (%) of grid-side current is proportional to the filter admittance at switching frequency. The linear equation of the TDD (%) by the filter admittance at switching frequency can be estimated as (28), regardless of the switching frequency.

$$TDD(\%) = 50R_{base} |Y_{filter}(j\omega_{sw})| \quad (28)$$

$$|Y_{filter}(j\omega_{sw})| = \frac{TDD_{target}(\%)}{50R_{base}} \quad (29)$$

Since the base impedance is fixed by specification of the inverter system, the filter admittance at switching frequency for target TDD (%) of grid-side current is designed as (29). As the TDD (%) of grid-side current is determined by the switching harmonics of grid-side current regardless of the filter type, the filter admittance at switching frequency, designed by (29) is applicable for any filter type.

2.4 Minimising current stress on inverter switch stack

The biggest difference between L filter and LCL filter is a filter capacitor. In aspect with grid-side current, the filter capacitor of LCL filter plays a role as a buffer to attenuate high-order switching harmonics of inverter output voltage. Unfortunately, in inverter side, filter capacitor of LCL filter can be considered as a capacitive load. The filter capacitor of LCL filter leads to raising switching ripple of inverter-side current which results in increasing stress on switch stack. To reduce the current stress on switch stack, switching ripple of inverter-side current should be limited.

One of authors here had investigated the relationship between a characteristic impedance of LC filter and inverter-side switching ripple in the inverter with LC filter. To minimise current stress of switch stack, the characteristic impedance of LC filter should be equal or larger than the base impedance of inverter system [25]. Here, the relationship between a characteristic impedance of LCL filter and inverter-side switching ripple is investigated just like LC filter. The characteristic impedance of LCL filter is defined as (30). Fig. 11 shows bode diagrams of LCL filters with different

Table 2 Specification of three-phase grid-connected inverter

Description	Symbol	Case I	Case II
Rated Power	P_{rate}	30kW	30kW
Grid voltage (line to line)	V_{LL_rms}	220Vrms	220Vrms
DC link voltage	V_{DC}	400V	400V
Grid frequency	f_g	60Hz	60Hz
Sampling frequency	f_{samp}	6kHz	20kHz
Switching frequency	f_{sw}	3kHz	10kHz

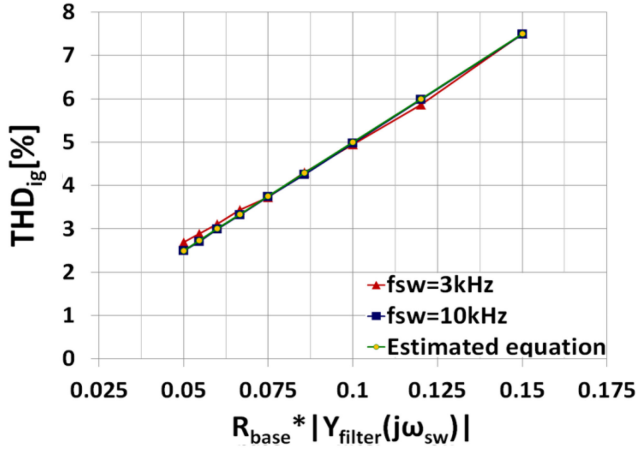


Fig. 10 THD of grid output currents in variation of the filter admittance at switching frequency

characteristic impedances and same admittance at switching frequency. When the characteristic impedance of LCL filter decreases in Fig. 11, the total filter inductance decreases and the filter capacitance increases.

$$Z_{LCL} = \sqrt{\frac{L}{C_f}} \quad (30)$$

Through MATLAB simulation, TDD (%) of grid-side current and the switching ripple factor of inverter-side current were investigated in variation of the characteristic impedance of LCL filter. The simulation conditions are shown in Case II of Table 2. LCL filters have the same admittance at switching frequency by (29), and the target TDD (%) of grid-side current is 5%. The ripple factor of inverter-side current and the ripple factor of grid-side current at a frequency or an angular speed are defined as follows.

$$RF_{if}(\omega) = \frac{I_f(\omega)}{I_{rate}} \times 100[\%] \quad (31)$$

$$RF_{ig}(\omega) = \frac{I_g(\omega)}{I_{rate}} \times 100[\%] \quad (32)$$

Fig. 12 shows TDD (%) of grid-side current and the ripple factor of inverter-side current at the switching frequency, according to per unit of the characteristic impedances of LCL filter over the base impedance. When the characteristic impedance of LCL filter is larger than the base impedance, TDD (%) of grid-side current keep below 5% and the ripple factor of inverter-side current at the switching frequency is limited under 10%. When the characteristic impedance of LCL filter is smaller than the base impedance, the ripple factors of inverter-side current at the switching frequency rise sharply and TDD (%) of grid-side current rise over 5%, even though the admittance of LCL filter at switching frequency are same.

Fig. 13 shows TDD (%) of grid-side current and the ripple factors of grid-side current at the resonance frequency, according to per unit of the characteristic impedances of LCL filter over the base impedance. When the characteristic impedance of LCL filter is

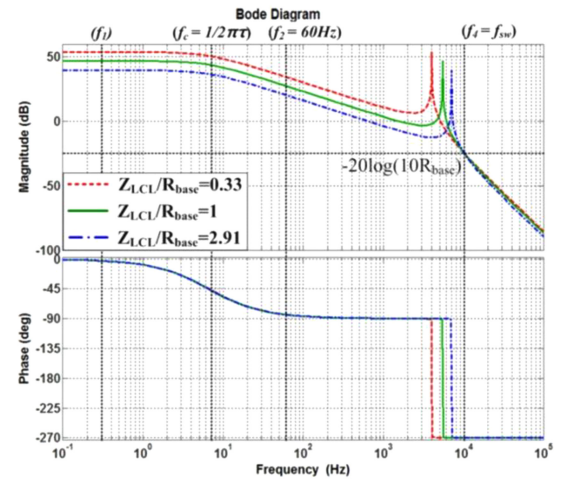


Fig. 11 Bode plots of LCL filters according to Z_{LCL}/R_{base} when the attenuating abilities at the switching frequency of them are equal

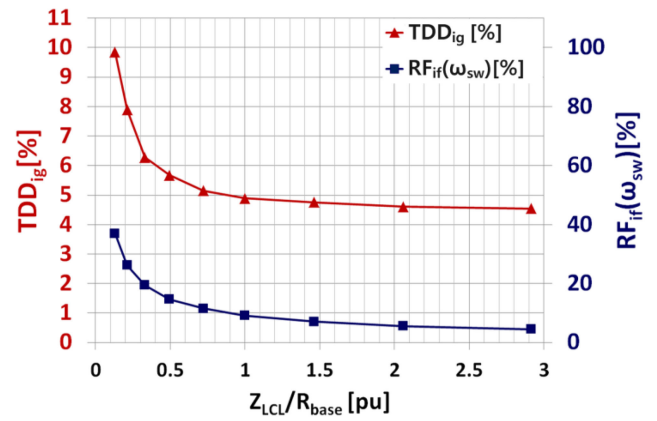


Fig. 12 TDD_{ig} and $RF_{if}(\omega_{sw})$ according to Z_{LCL}/R_{base}

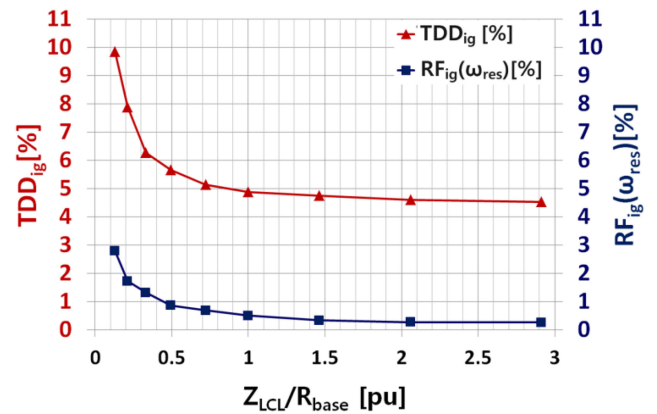


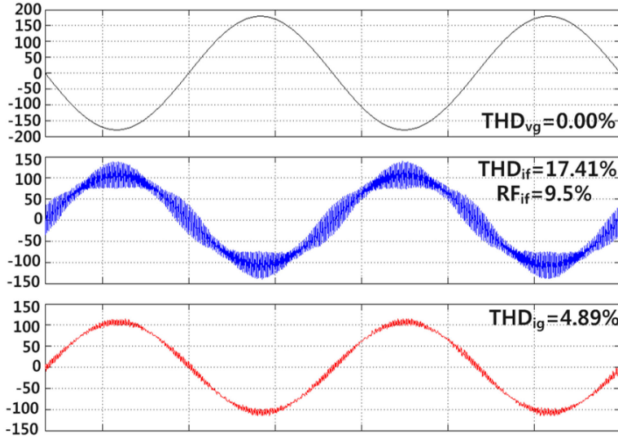
Fig. 13 TDD_{ig} and $RF_{ig}(\omega_{res_LCL})$ according to Z_{LCL}/R_{base}

larger than the base impedance, TDD (%) of grid-side current keep below 5% and the ripple factors of grid-side current at the resonance frequency is limited under 0.5%. When the characteristic impedance of LCL filter is smaller than the base impedance, the ripple factor of grid-side current at the resonance frequency rise sharply and TDD (%) of grid-side current rise over 5%, even though the admittance of LCL filter at switching frequency are same.

Therefore, in order to achieve the target TDD (%) of grid-side current by the admittance of LCL filter at the switching frequency in (29) and to limit the ripple factor of inverter-side current at the switching frequency under 10%, the characteristic impedance of LCL filter should be equal or larger than the base impedance of the inverter system.

Table 3 Elements of LCL filter by proposed design method

Description	Symbol	Value
Base impedance	R_{base}	1.613Ω
Target TDD of grid-side current	TDD_{target}	5%
Inverter-side filter inductance	L_f	81.5uH
Grid-side filter inductance	L_g	27.2uH
Filter capacitance	C_f	41.8uF

**Fig. 14** Simulated waveforms of V_g , i_f and i_g by proposed LCL design method

2.5 Proposed LCL filter design method

This paper establishes three simultaneous equations by analysing filter inductance ratio, filter admittance at switching frequency, and characteristic impedance for optimal LCL filter design.

1. $r = \frac{L_f}{L} = 0.75$
2. $2|Y(j\omega_{sw})_{LCL}| = \frac{(1/(r(1-r)L^2C_f))}{\omega_{sw}^2 \left(\omega_{sw}^2 - \frac{1}{r(1-r)LC_f} \right)} = \frac{TDD_{target}(\%)}{50R_{base}}$
3. $3Z_{LCL} = \sqrt{L/C_f} = R_{base}$

First equation means the optimal inductance ratio to minimise total filter inductance. Second equation means the design equation of filter admittance at switching frequency to meet target TDD (%) of grid-side current. Final equation means the characteristic impedance of LCL filter to minimise current stress on switch stack and to limit grid-side resonance current around the resonance frequency. If first and third equations are substituted for second equation, the design equation for the total filter inductance can be expressed as (33).

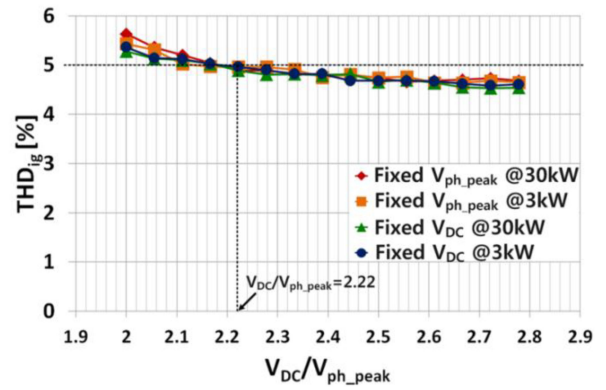
$$\frac{1}{L^3} + \frac{\omega_{sw} TDD_{target}(\%)}{50R_{base}} \frac{1}{L^2} - \frac{0.1875 TDD_{target}(\%)}{50R_{base}^3} = 0 \quad (33)$$

In the coefficient condition of (33), the solutions have a positive real root and two imaginary roots. The formula of cubic equation can make (33) solved. As a result, LCL filter elements to satisfy three goals can be uniquely designed without try & error.

3 Evaluation of proposed design method For LCL filter

3.1 Simulation evaluation of proposed design method

To evaluate the proposed design method, MATLAB simulation was performed. A simulation condition follows Case II of Table 2. The filter elements designed by the proposed method in the condition are listed in Table 3. Fig. 14 shows simulated waveforms by the designed LCL filter: grid-phase voltage, inverter-side current, and grid-side current. In the designed filter, the simulation result

**Fig. 15** THD of grid-side currents in variation of V_{DC} , V_{ph_peak} and P_{rate}

verifies that TDD (%) of grid-side current is satisfied with the grid regulation and the ripple factor of inverter-side current at switching frequency is limited below 10%.

Fig. 15 shows THD of grid-side current by proposed filter design method in the variation of V_{DC} , V_{ph_peak} and P_{rate} , where fixed V_{ph_peak} is 180 V and fixed V_{DC} is 400 V. Generally, a peak phase voltage in the three-phase PWM inverter is defined as (34).

$$V_{ph_peak(1)} = \frac{V_{DC}}{2} m_a \quad (34)$$

where $V_{ph_peak(1)}$ is the peak-phase voltage in the three-phase PWM inverter, V_{DC} is DC source voltage, and m_a is an amplitude modulation index. The value of m_a is between 0 and 1. If the phase voltage of grid-connected inverter is equal to the grid-phase voltage, DC source voltage of the inverter should be at least twice bigger than peak-phase voltage of grid. When the ratio of DC source voltage over peak-phase voltage of grid is smaller than 2.22, it is shown that TDD (%) of grid-side current by the proposed LCL filter increases above the target TDD (%). When the ratio of DC source voltage over peak-phase voltage of grid is larger than 2.22, it is shown that TDD (%) of grid-side current by the proposed LCL filter decreases below the target TDD (%). It physically means that the proposed design method is effective, in operating conditions that the amplitude modulation index m_a is under 90%.

3.2 Experimental evaluation of proposed design method

To evaluate the proposed design method, an experiment was performed in the inverter specification of Table 4. The designed filter parameters and the manufactured filter parameters are listed in Table 5. There are some differences between the designed filter parameters and manufactured ones because of manufacturing condition in the market. The differences in filter inductances are 3% and that of filter capacitance is 7%. However, those differences are within the tolerance levels of the manufactured filter reactors and capacitance which is 10%.

Fig. 16 shows experimental waveforms by the designed LCL filter. THD of waveform is considered up to 50th-order harmonics, followed in IEEE 519-2014. First upper level waveform is a grid-phase voltage: $THD_{Vg} = 1.22\%$. Second upper level waveform is a phase voltage of the filter capacitor: $THD_{Vc} = 1.53\%$. Third upper level waveform is an inverter-side current: $THD_{If} = 4.80\%$. Lowest level waveform is a grid-side current: $THD_{Ig} = 4.63\%$.

Fig. 17 shows experimental FFT analysis of grid-side current by the designed LCL filter. Even though a resonance frequency by the designed LCL filter is 5205 Hz in Table 5, a resonance frequency in the FFT analysis of Fig. 17 appears at about 3900 Hz. It is considered that the change of resonance frequency in Fig. 17 and the reduction of switching ripple of inverter side current in Fig. 16 are caused by the additional grid impedance of a transformer of 5kVA capacity, installed at the point of common coupling (PCC).

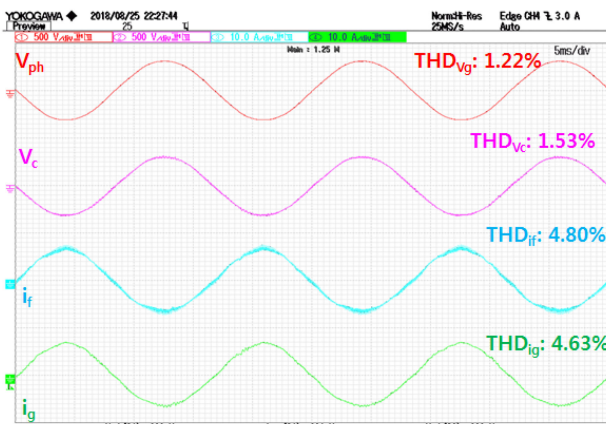
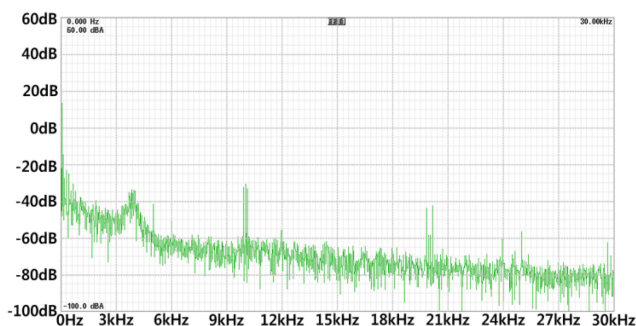
From the change of resonance frequency on the LCL filter, grid impedance of the transformer can be estimated. The additional inductance is about 1.1mH. Fig. 18 shows simulated waveform of

Table 4 Specification of three phase grid-connected inverter

Description	Symbol	Value
Rated Power	P_{rate}	30kW
Grid voltage (line to line)	V_{LL_rms}	380Vrms
Base impedance	R_{base}	48.13 Ω
DC link voltage	V_{DC}	700V
Grid frequency	f_g	60Hz
Sampling frequency	f_{samp}	20kHz
Switching frequency	f_{sw}	10kHz
%dead time	T_{dead}/T_{sw}	2.5%
Target TDD of grid-side current	TDD_{target}	5%

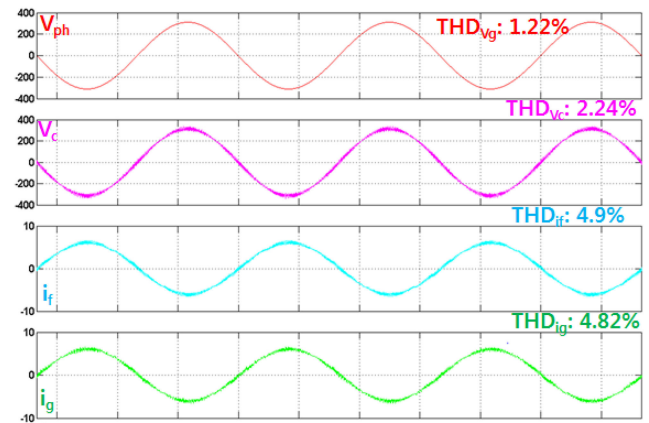
Table 5 Elements of LCL filter by proposed design method

Description	Designedvalue	Manufacturedvalue
Inverter-side filter inductance	2.43mH	2.5mH
Grid-side filter inductance	0.81mH	0.8mH
Filter capacitance	1.4uF	1.5uF
Resonance frequency	5270Hz	5205uF

**Fig. 16** Experimental waveform of V_g , V_c , i_f and i_g by the designed LCL filter**Fig. 17** Experimental FFT analysis of grid-side current by the designed LCL filter

V_g , V_c , i_f and i_g under same conditions with the experiment. The simulation waveform is considered up to 50th-order harmonics equally with experiment waveform. Table 6 shows THD value of V_g , V_c , i_f and i_g in experiment and simulation. From Table 6, it is confirmed that experimental result and simulated result are similar and it is proved that TDD (%) of grid-side current is successfully limited below 5% by the proposed design method.

Therefore in experiment and simulation, even though resonance frequency in LCL filter is changed by unpredictable grid impedance and low-order harmonics are generated by dead time control, it is verified that TDD(%) of grid-side current is successfully limited below 5% by the proposed design method.

**Fig. 18** Simulated waveform of V_g , V_c , i_f and i_g under same condition with the experiment**Table 6** THD value of V_g , V_c , i_f and i_g in experiment and simulation

Symbol	Experimentalvalue	Simulatedvalue
THD_{Vg}	1.22%	1.22%
THD_{Vc}	1.53%	2.24%
THD_{if}	4.8%	4.9%
THD_{ig}	4.63%	4.82%

4 Conclusion

In this study, three design goals are selected for LCL filter: filter inductance ratio to minimise total filter inductance, filter admittance to meet grid regulation, and characteristic impedance for low current stress of switch stack. Through analysing frequency responses in LCL filter, the optimal inductance ratio is discovered that the inverter-side inductance should be three times as large as grid-side inductance. Also, for a target TDD (%) of grid-side current, the equation to estimate filter admittance at switching frequency was suggested. When the characteristic impedance of LCL filter is larger than the base impedance, the ripple factor of inverter-side current at the switching frequency is limited under 10%. Here, third-order optimal design equation is proposed to meet the three design goals. The proposed method can solve unique filter elements for LCL filter without iterative try & error.

The proposed design method is evaluated by simulations and an experiment. In the simulations, those are verified that the switching ripple of inverter-side current and TDD of grid-side current are limited below target values and that the proposed method can be applied in variations of DC source voltage and grid voltage. In the experiment and simulation, despite the low-order harmonics by dead time control and the change of resonance frequency in LCL filter by unpredictable grid impedance, it is verified that TDD (%) of grid-side current is successfully limited below 5% by the proposed design method.

5 Acknowledgments

This research was supported by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education (2016R1D1A3B01008279).

6 References

- [1] Wang, G., Konstantinou, G., Townsend, C.D., *et al.*: 'A review of power electronics for grid connection of utility-scale battery energy storage systems', *IEEE Trans. Sustainable Energy*, 2016, 7, (4), pp. 1778–1790
- [2] IEEE Standard 519-2014 'IEEE Recommended Practice and Requirements for Harmonic Control in Electric Power Systems', Transmission and Distribution Committee.
- [3] Tan, Y., Yao, W., Loh, P.C., *et al.*: 'Design of LCL filters with LCL resonance frequencies beyond the nyquist frequency for grid-connected converters', *IEEE Trans. Power Electron*, 2016, 4, (1), pp. 3–14

- [4] Liserre, M., Blaabjerg, F., Hansen, S.: 'Design and control of an LCL-filter-based three-phase active rectifier', *IEEE Trans. Ind. Appl.*, 2005, **41**, (15), pp. 1281–1291
- [5] Gabe, I.J., Montagner, V.F., Pinheiro, H.: 'Design and implementation of a robust current controller for VSI connected to the grid through an LCL filter', *IEEE Trans. Power Electron.*, 2009, **24**, (6), pp. 1444–1452
- [6] Peña-Alzola, R., Liserre, M., Blaabjerg, F., *et al.*: 'Analysis of the passive damping losses in LCL-filter-based grid converters', *IEEE Trans. Power Electron.*, 2013, **28**, (6), pp. 2642–2646
- [7] Beres, R.N., Wang, X., Blaabjerg, F., *et al.*: 'Optimal design of high-order passive-damped filters for grid-connected applications', *IEEE Trans. Power Electron.*, 2016, **31**, (3), pp. 2083–2098
- [8] Pan, D., Ruan, X., Bao, C., *et al.*: 'Capacitor-current-Feedback active damping with reduced computation delay for improving robustness of LCL-type grid-connected inverter', *IEEE Trans. Power Electron.*, 2014, **29**, (7), pp. 3414–3427
- [9] Bao, C., Ruan, X., Wang, X., *et al.*: 'Step-by-step controller design for LCL-type grid-connected inverter with capacitor-current-feedback active-damping', *IEEE Trans. Power Electron.*, 2014, **29**, (3), pp. 1239–1253-14
- [10] Wagner, M., Barth, T., Alvarez, R., *et al.*: 'Discrete-time active damping of LCL -resonance by proportional capacitor current feedback', *IEEE Trans. Ind. Appl.*, 2014, **50**, (6), pp. 3911–3920
- [11] Xin, Z., Loh, P.C., Wang, X., *et al.*: 'Highly accurate derivatives for LCL-filtered grid converter with capacitor voltage active damping', *IEEE Trans. Power Electron.*, 2016, **31**, (5), pp. 3612–3625
- [12] Tang, Y., Loh, P.C., Wang, P., *et al.*: 'Exploring inherent damping characteristic of LCL-filters for three-phase grid-connected voltage source inverters', *IEEE Trans. Power Electron.*, 2012, **27**, (3), pp. 1433–1443
- [13] Wang, X., Blaabjerg, F., Loh, P.C.: 'Grid-current-feedback active damping for LCL resonance in grid-connected voltage-source converters', *IEEE Trans. Power Electron.*, 2016, **31**, (1), pp. 213–223
- [14] Malinowski, M., Bernet, S.: 'A simple voltage sensorless active damping scheme for three-phase PWM converters with an LCL filter', *IEEE Trans. Power Electron.*, 2008, **55**, (4), pp. 1876–1880
- [15] Miskovic, V., Blasko, V., Jahns, T.M., *et al.*: 'Observer-Based active damping of LCL resonance in grid-connected voltage source converters', *IEEE Trans. Ind. Appl.*, 2014, **50**, (6), pp. 3977–3985
- [16] Muhlethaler, J., Schweizer, M., Blattmann, R., *et al.*: 'Optimal design of LCL harmonic filters for three-phase PFC rectifiers', *IEEE Trans. Power Electronics*, 2013, **28**, (7), pp. 3114–3125
- [17] Bueno, E.J., Cóbrecas, S., Rodríguez, F.J., *et al.*: 'Design of a back-to-back NPC converter interface for wind turbines with squirrel-cage induction generator', *IEEE Trans. Energy Convers*, 2008, **23**, (3), pp. 932–945
- [18] Jiao, Y., Lee, F.C.: 'LCL filter design and inductor current ripple analysis for a three-level NPC grid interface converter', *IEEE Trans. Power Electron.*, 2015, **30**, (9), pp. 4659–4668
- [19] Channegowda, P., John, V.: 'Filter optimization for grid interactive voltage source inverters', *IEEE Trans. Ind. Electron.*, 2010, **57**, (12), pp. 4106–4114
- [20] Kouchaki, A., Nymand, M.: 'Analytical design of passive LCL filter for three-phase two-level power factor correction rectifiers', *IEEE Trans. Power Electron.*, 2018, **33**, (4), pp. 3012–3022
- [21] Tang, Y., Yao, W., Loh, P.C., *et al.*: 'Design of LCL filters with LCL resonance frequencies beyond the nyquist frequency for grid-connected converters', *IEEE J. Emerg. Sel. Top. Power Electron.*, 2016, **4**, (1), pp. 3–14
- [22] Reznik, A., Simões, M.G., Al-Durra, A., *et al.*: 'LCL filter design and performance analysis for grid-interconnected systems', *IEEE Trans. Ind. Appl.*, 2014, **50**, (2), pp. 1225–1232
- [23] Xu, J., Xie, S., Kan, J.: 'LCL-filter design for grid-connected inverter to suppress grid-induced low-order current harmonics', *Proc. IEEE Energy Convers. Congr. Exposition (ECCE)*, Montreal, QC, Canada, 2015, pp. 1178–1183
- [24] Jiao, Y., Lee, F.C.: 'LCL filter design and inductor current ripple analysis for a three-level NPC grid interface converter', *IEEE Trans. Power Electron.*, 2015, **30**, (9), pp. 4659–4668
- [25] Kim, H.S., Sul, S.K.: 'A novel filter design for output LC filters of PWM inverters', *J. Power Electron.*, 2011, **11**, (1), pp. 74–81